

Question	Marking Guidance	Mark	Comments
10.1	1. Fat (store) used in respiration/metabolism; 2. Less energy/food (store) is required due to low respiration/metabolism OR Less energy/food (store) is required due less movement; 3. Gluconeogenesis; 4. Low surface area to volume reduces heat loss OR Fat (layer/insulation) reduces heat loss; 5. Long loop of Henle so less water lost; 6. Water provided from respiration; 7. Reduced/no urination; 8. Less evaporation;	3 max	Mark points 1 to 4 = 2 max. Mark points 5 to 8 = 2 max. 1 and 2. Reject respiration 'uses energy' or 'produces energy'. 3. Accept description in terms of using glycerol, fatty acids or amino acids. 5. Accept thick medulla (in kidney) for long loop of Henle. 8. Accept less sweating.
10.2	1. (Lower metabolism so) less/low CO₂ (in blood); 2. (Detected by) chemoreceptors; 3. (Chemoreceptors) located in aorta/medulla OR (Chemoreceptors) located in carotid artery; 4. Fewer impulses to cardiac centre; OR Fewer impulses to medulla (oblongata); 5. (More) impulses along parasympathetic/vagus pathway/neurones/nerve OR Fewer impulses along sympathetic pathway/neurones/nerve; 6. (To) SAN;	4 max	If neither mark point 1 or 2 credited = 3 max. 1. Accept increase in pH or decrease in H ions/acidity for less CO₂. Ignore baroreceptors. 2. Ignore detects oxygen, (concentration). 3. Accept carotid body or aortic body. 4 and 5. Reject (once only) reference to 'an/one impulse'. 4 and 5. Reject 'signals', 'messages' (once only) for 'impulses' 4 and 5. Accept 'action potential/s' for impulses.

10.3	1. Allow passage of protons/H ⁺ ; 2. (Energy) released as heat;	2	1. Ignore direction of movement/diffusion/active transport. 2. Accept 'produces heat' but reject 'produces 'heat energy'.
10.4	1. Less snow so less camouflage; 2. More hares seen/eaten/killed by predators;	2	1. Accept 'snow melts' 1. Accept description of less camouflage, e.g. more hares seen. 2. Accept description of predation.
10.5	1. Hares which moult earlier (more likely to) survive; 2. Hares which moult earlier (more likely to) reproduce; 3. Pass on (advantageous) <u>allele</u> ; 4. Frequency of <u>allele</u> increases (in future populations);	4	1. Accept less likely to be killed for 'survive'. 1. Accept description of survival e.g. not killed/eaten. 1. Accept moult quicker/faster for earlier. 1 and 2. Answers must be in the context of moulting earlier/quicker/faster. Accept rabbits for hares. 2 and 3. Accept 'pass on allele to offspring' or 'to next generation' = 2 marks. 4. 'More alleles' is not enough for a mark.